

```

5 def filter_data(
6     df: pl.DataFrame,
7 ) -> pl.DataFrame:
8
9     df = df.filter(
10         (pl.col("defense_system_cluster_id") == pl.col("cluster_id")) &
11         (pl.col("Organism") == "Homo sapiens (Human)")
12     )
13     logger.info(f"N rows after filtering: {len(df)}")
14
15     return df

```

```

17 # What are the species annotations?
18 print("Unique values for species:")
19 print(df_def_cleaned["Organism"].unique())

```

```

21 df_def_filtered = filter_data(
22     df=df_def_cleaned,
23 )

```

```

25 df_def_filtered.write_csv(
26     f"{args['out_path']}.csv",
27     separator="\t",
28     include_header=True
29 )

```

```

TCAG
TCAGT
TCAGTC
GTCAGTCA
TCAGTCA
CAGTCAG
AGTCAG
TCAGTC
TCAGTC
CAGTCAG
AGTCAGTC
TCAGTCAGTCAGTCAG
GTCAGTCAGTCAG
GTCAGTCAG
GTCA TCAGT
AGTCA CAGTCA
CAGTCAG AGTCAGTC
CAGTCAGT TCAGTCA
AGTCAGT TCAGTCAG
AGTCAGTC CAGTCAGT
GTCAGTCA GTCAGTCA
GTCAGTCAG TCAGTCAG
CAGTCAG AGTCAGT
CAGTCAGT GTCAGTC
AGTCAGT CAGTCAG
TCAGTCA CAGTCAGT
CAGTCAG AGTCAGTC
AGTCAG
GTCAGTC
TCAGTC
AGTCAG
TCAG

```

```

TCAG
AGTCAG
GTCAGT
TCAGTCA
CAGTCAG
AGTCAGT
TCAGTC
TCAGTC
CAGTCAG
CAGTCA
TCAGTCAGTCAGTCAG
GTCAGTCAGTCAG
GTCAGTCAG
TCAGT
CAGTCA
AGTCAGTC
AGTCAGTC
TCAGTCA
AGTCAGT
TCAGTCAG
CAGTCAGT
GTCAGTC
CAGTCAG
CAGTCAGT
AGTCAGTC
TCAG

```

Bacterial anti-phage structural homology: update 1

Julian Moran

2026-02-19

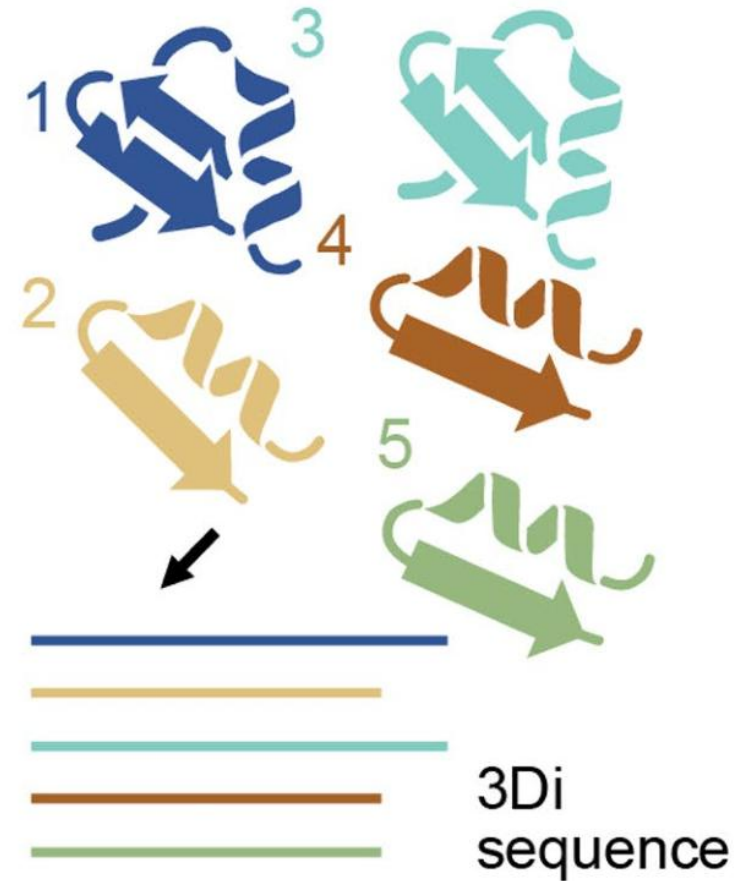
The Centre for Applied Genomics, SickKids

Background

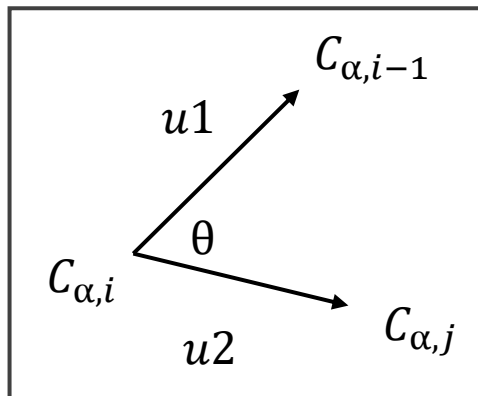
Foldseek, a protein structure search algorithm, depends on **3Di**, a method for representing 3D folded protein structures as one-dimensional sequences of a 20-letter code.¹

Each character in the 20-letter code is represented in the latent space between an autoencoder and decoder neural network.

The autoencoder receives as input a 10-dimensional feature vector, where the features encapsulate the spatial relationship between two nearest spatial neighbor residues, i and j , in the 3D structure.



Adapted from Barrio-Hernandez et al. 2023.¹



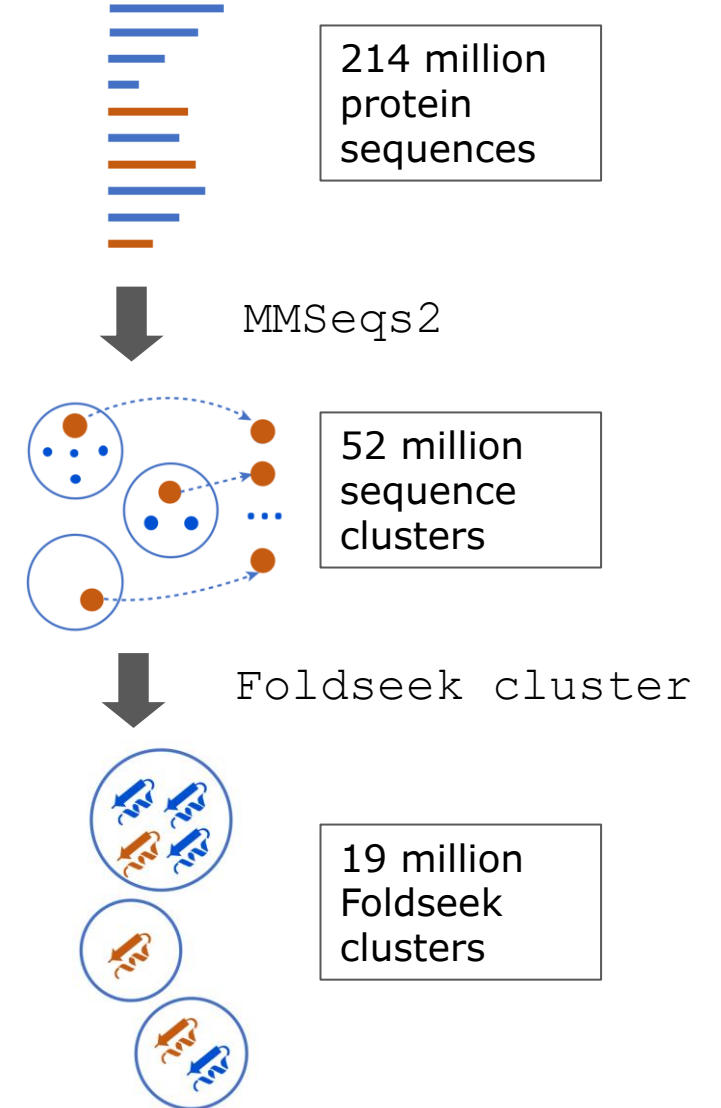
$$\cos(\theta_{u1,u2})$$

Background

Foldseek cluster, a clustering algorithm, leverages both primary amino acid sequence homology and structural homology to group proteins together.²

It takes as input the entire AlphaFold database (214 million sequences and predicted structures) clustered with MMSeqs2, which uses primary amino acid sequence. Here, sequence cluster membership is thresholded at 50% sequence identity and 90% sequence alignment overlap, yielding 52 million clusters.

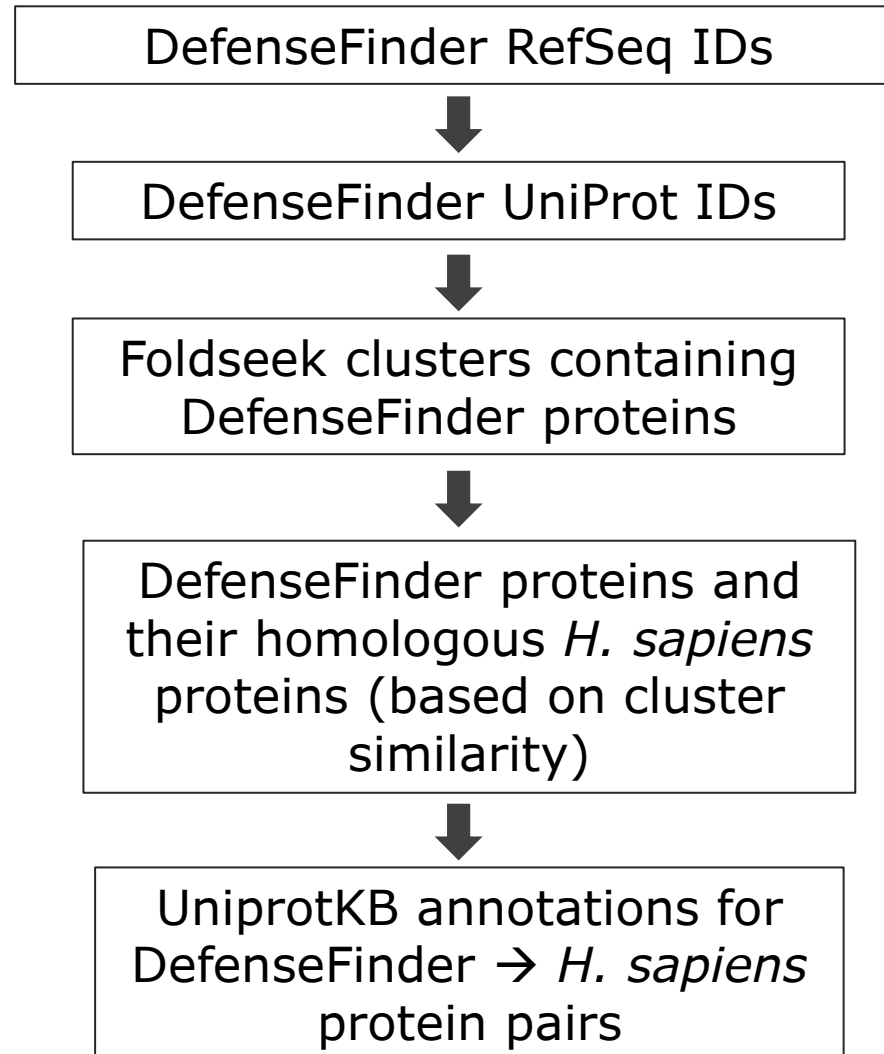
Foldseek cluster then further clusters the 3Di sequences of "representative structures" (●) from each cluster (i.e. structure with highest pLDDT), with membership threshold at e-value 0.01 and 90% structural alignment overlap, yielding 19 million clusters.



Research questions

1. Which human proteins in the Foldseek cluster dataset exhibit strong homology with bacterial immune system proteins?
2. Based on the structural homology, can we hypothesise about previously uncharacterised functions of the human proteins?

A Jain pipeline summary



Anjali Jain

Bioinformatician
at the Centre for
Computational
Medicine (CCM)

321,575 protein pairs
for the Zorya, Vipirin,
and Gasdermin bacterial
immune systems

Downstream prioritization

321,575

DefenseFinder → *H. sapiens*
protein pairs



`DefenseFinder cluster_id ==
H. sapiens cluster_id`

279

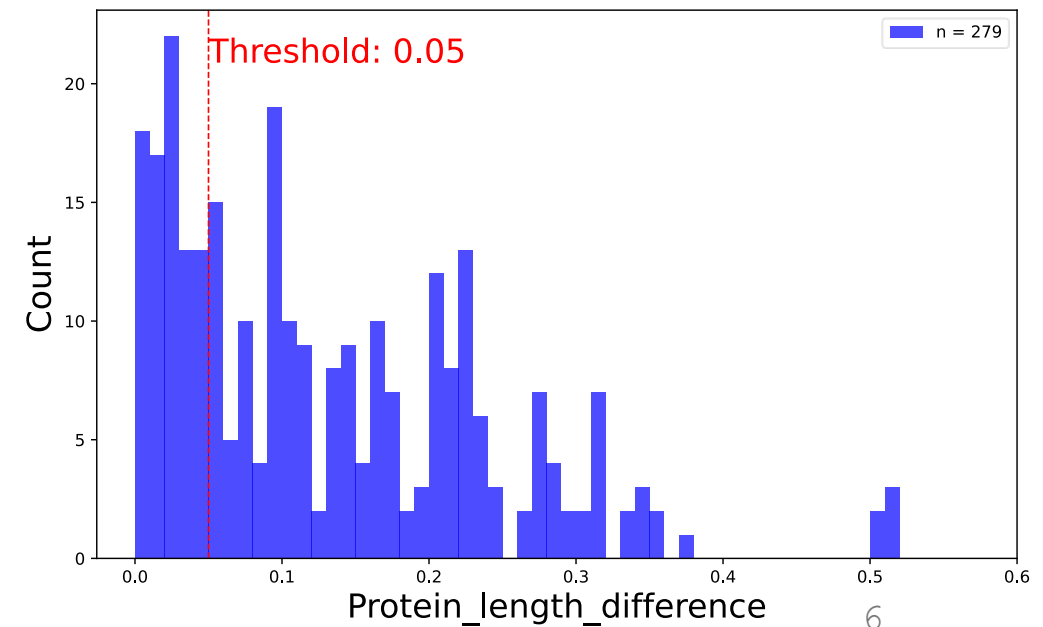
DefenseFinder → *H. sapiens*
protein pairs



`Protein_length_difference ≤ 0.05`

34

DefenseFinder → *H. sapiens*
protein pairs



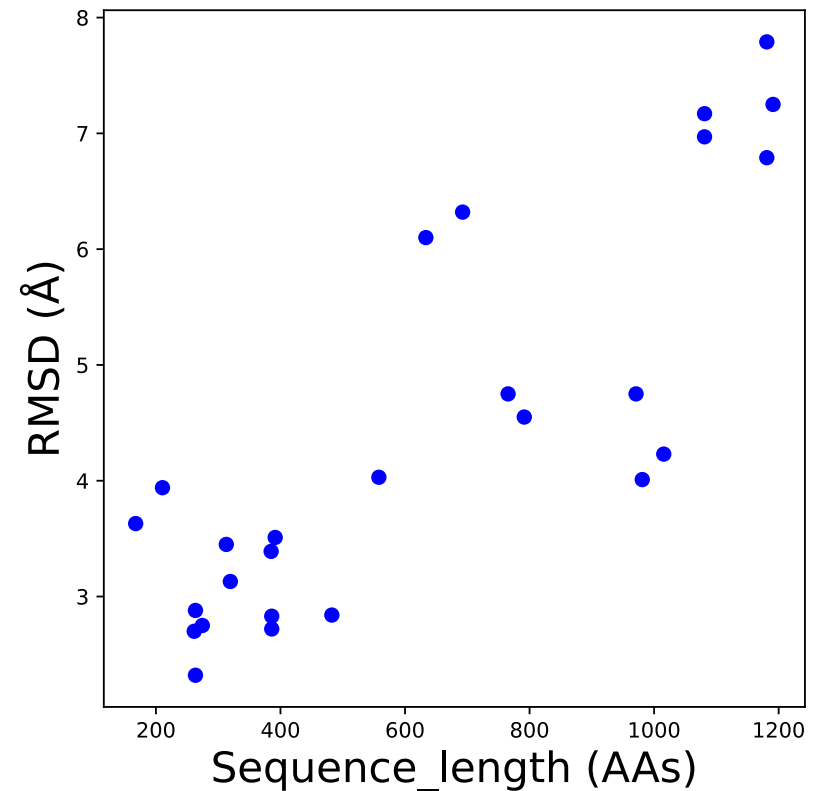
279

DefenseFinder → *H. sapiens*
same-cluster protein pairs

contains proteins from 17
distinct clusters

functions generally
pertain to triphosphate
metabolism and chromosome
organization

G profiler GO terms	
name	p_value
ATP hydrolysis activity	5.32E-14
ATP-dependent activity	8.18E-12
ribonucleoside triphosphate phosphatase activity	5.90E-10
pyrophosphatase activity	1.10E-09
hydrolase activity, acting on acid anhydrides	1.11E-09
hydrolase activity, acting on acid anhydrides, in phosphorus-containing anhydrides	1.11E-09
condensed chromosome	7.14E-09
condensed nuclear chromosome	8.75E-09
ATP binding	1.13E-08
chromosome organization	1.19E-08



**cDNA FLJ56591, highly similar to
Eukaryotic translation initiation factor
2C3**

Molecular functions

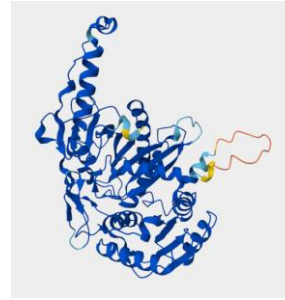
- DNA binding
- DNA-binding transcription factor activ.
- RNA binding

Biological processes

- immune system process
- carbohydrate metabolic process
- DNA metabolic process

Domains

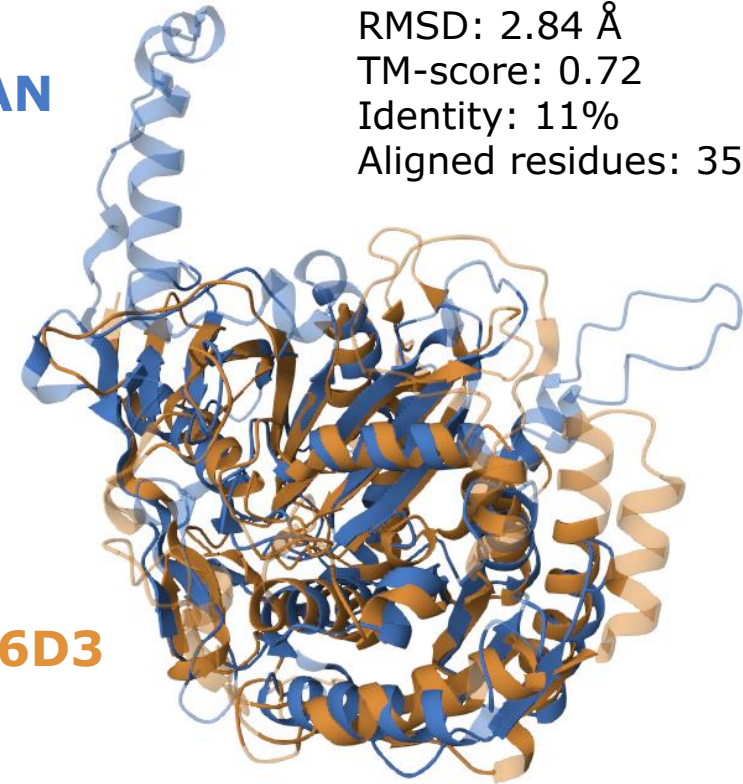
- Argonaute-like, PIWI domain (29-454)
- Ribonuclease H-like superfamily (140-456)



- Very high (pLDDT > 90)
- Confident (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very low (pLDDT < 50)

B4DY67_HUMAN
(496 AAs)

RMSD: 2.84 Å
TM-score: 0.72
Identity: 11%
Aligned residues: 351



Protein argonaute (Bacteroides sp. AM16-24)

Domains

- Ribonuclease H-like superfamily (2-469)
- PIWI domain (144-457)

A0A414Z6D3
(469 AAs)

Radical S-adenosyl methionine domain containing 2

Molecular functions

- Small molecule binding

Domains

- RSAD2 and Vipirin antiviral (47-239)

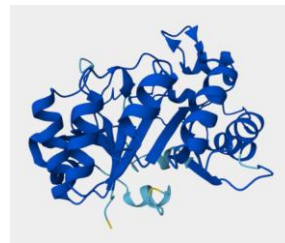
S-adenosylmethionine-dependent nucleotide hydratase (Candidatus Thiodictyon syntrophicum)

Molecular functions

- Catalytic activity
- Small molecule binding

Biological functions

- response to stimulus

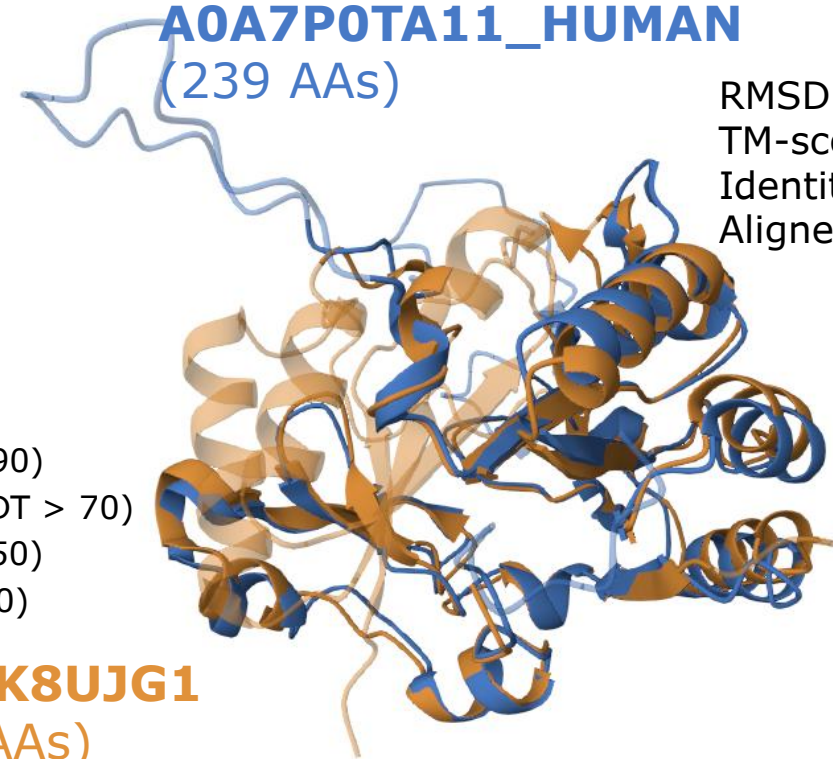


- Very high (pLDDT > 90)
- Confident (90 > pLDDT > 70)
- Low (70 > pLDDT > 50)
- Very low (pLDDT < 50)

A0A7P0TA11_HUMAN (239 AAs)

RMSD: 2.70 Å
TM-score: 0.64
Identity: 32%
Aligned residues: 186

A0A2K8UJG1 (284 AAs)



Next steps

1. Automate pairwise structural alignment with PyMol
 - domain-specific alignment (likely more biologically interesting)
 - overlay domain annotations
2. Prioritize on basis of alignment properties
 - do we ever align to H. sapiens protein domains with poorly characterised function?
 - conversely, are domains that do not align in H. sapiens poorly characterised?

References

1. Van Kempen M, et al. Fast and accurate protein structure search with Foldseek. *Nat Biotechnol.* 2024;42:243-246. DOI: 10.1038/s41587-023-01773-0
2. Barrio-Hernandez I, et al. Clustering predicted protein structures at the scale of the known protein universe. *Nature.* 2023;622:637-645. DOI: 10.1038/s41586-023-06510-w